

# BeeTrust AI



## Making Honey More Transparent , Trusted & Smarter

Team ID: [ SIH26\_241]  
Team Name: [ Team Insite]

Domain : Agriculture & Rural Development  
Category: Software Edition

**“From the Beekeeper’s Hive to the Consumer’s Home.”**

# Problem Statement

## The Core Bottleneck

The traditional honey supply chain operates as a blind, unverified loop. A lack of transparent tracking damages consumer confidence, while a lack of data isolates rural producers, directly restricting KVIC's national "Honey Mission" from scaling.

## Why This Matters: The Triple-Impact Crisis



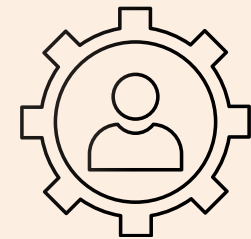
### The Consumer Trust Deficit Market Impact

**The Vulnerability:** Consumers have limited access to transparent, verifiable information about a honey product's origin, harvest history, and purity testing. [11] [12]  
**The Economic Consequence:** Honey adulteration can damage consumer confidence and hurt genuine beekeepers, while better traceability can improve trust and willingness to pay for authentic products. [10] [11] [13]



### Rural Livelihood Stagnation Socio-Economic Impact

**The Vulnerability:** Small-scale beekeepers may not have access to affordable monitoring and diagnostic tools, making it harder to detect colony stress and diseases early. [14] [15]  
**The Economic Consequence:** Better monitoring and early detection can help reduce avoidable colony losses and improve beekeeping productivity, while KVIC's Honey Mission aims to support employment and increase rural incomes. [14] [17] [18]



### Institutional Fragmentation Administrative Impact

**The Vulnerability:** Limited real-time data from local beekeepers makes it difficult for KVIC to track production and identify problems before they spread. [19] [20]  
**The Economic Consequence:** A unified digital platform can connect verified beekeeper records with production and hive data, enabling KVIC to identify regional problems earlier and make more informed decisions. [14] [20]

## The Pivot: Why Our Software Simulation Approach Works

By replacing expensive physical hardware with a Deterministic Python Simulation Engine, we lower the technology entry barrier for rural cooperatives to zero cost. We deliver the analytical benefits of high-end smart hives directly to the lowest-income beekeepers via an offline-first mobile framework



# Literature survey and Existing Solutions

| WHAT EXISTS TODAY?                                                                                                                                                                | WHAT DO THEY HELP WITH?                                                                                             | WHERE IS THE GAP?                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Honey testing &amp; certification</b> — Honey is tested in laboratories to check its quality, purity and possible adulteration. [1]                                            | Helps make sure the honey meets <b>quality and safety standards</b> before reaching the market.                     | ✗ Consumers still don't always get easy access to the <b>testing history of the specific batch</b> they are buying. [7] [11]              |
| <b>Digital beekeeper registration &amp; traceability</b> — Government initiatives like Madhukranti digitally register beekeepers and help trace honey back to its source. [1] [3] | Helps identify <b>where the honey comes from</b> and makes basic traceability easier.                               | ✗ The complete journey from <b>beekeeper → hive → lab → product → consumer</b> isn't available in one place. [1] [7]                      |
| <b>IoT-based hive monitoring</b> — Sensors can monitor things like hive temperature, humidity and weight. [9] [14] [20]                                                           | Helps beekeepers <b>keep an eye on their hives</b> remotely and notice unusual changes sooner.                      | ✗ This hive data isn't connected with <b>lab results, beekeeper records and consumer information</b> in one system. [7]                   |
| <b>Blockchain-based traceability</b> — Blockchain has been explored for tracking honey through different stages of the supply chain. [5] [6] [7]                                  | Helps create records that are harder to alter and makes the movement of honey easier to track.                      | ✗ Most existing solutions focus on individual parts of the problem, rather than one platform covering the entire honey ecosystem. [5] [7] |
| <b>KVIC &amp; government support programmes</b> — Programmes such as Honey Mission provide beekeepers with training, equipment and technical support. [17] [18]                   | Helps small beekeepers <b>start, manage and improve their beekeeping activities</b> and supports rural livelihoods. | ✗ Beekeepers could benefit from more accessible digital tools for monitoring, diagnosis and record keeping. [17]                          |
| KVIC Honey MIS & digital reporting — KVIC already uses digital systems to collect and manage honey-sector information. [19]                                                       | Helps with <b>organising programme and sector-level information</b> digitally.                                      | ✗ Information from <b>beekeepers, hives, laboratories and consumers</b> isn't brought together into one platform. [19] [1]                |

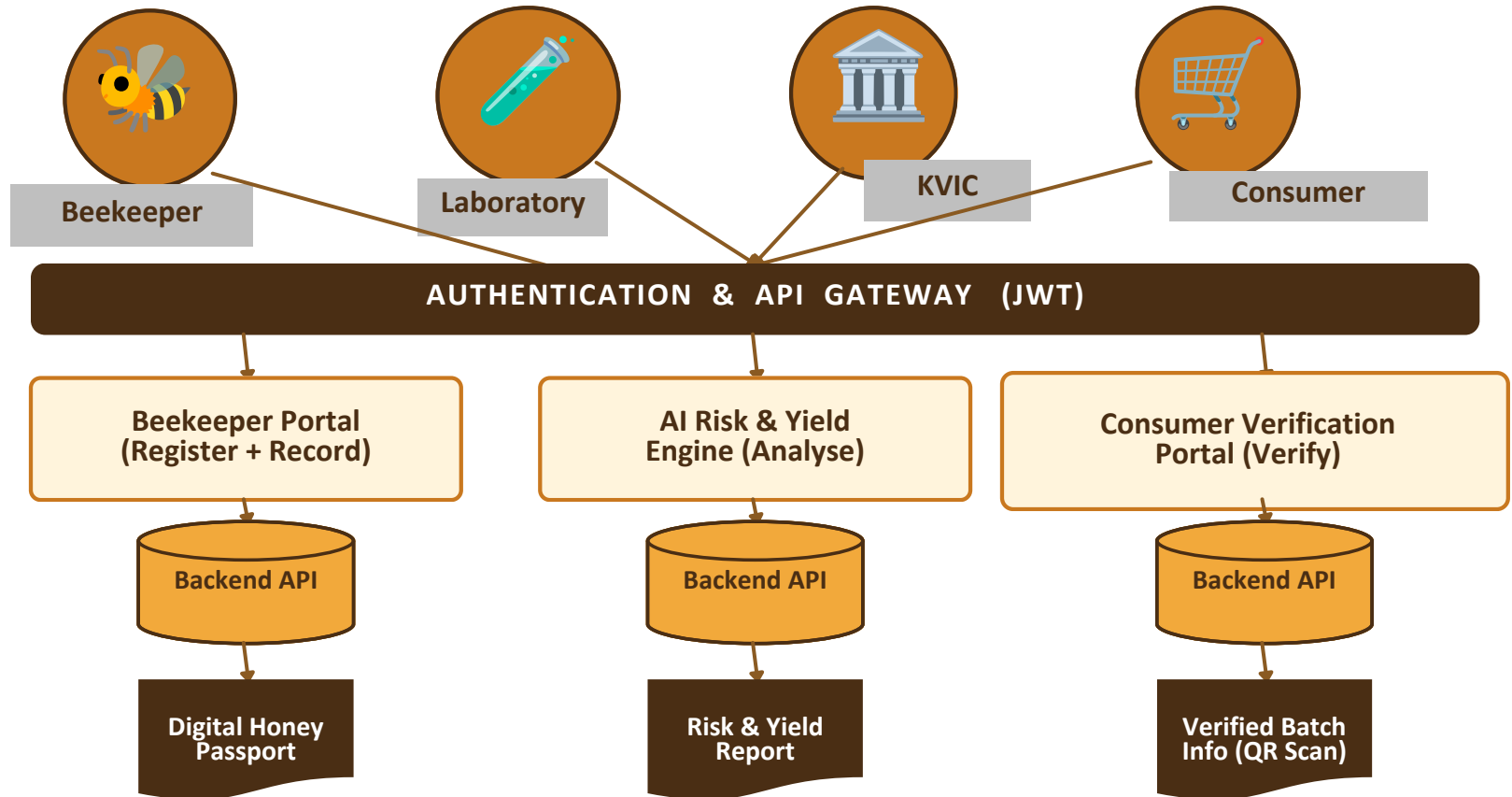
## OUR OPPORTUNITY

Build one simple digital platform that connects the different stakeholders and makes honey information easier to record, verify and understand.

BeeTrust AI connects beekeepers, laboratories, KVIC and consumers through one digital record per honey batch. A Digital Honey Passport gives every batch a unique identity — origin, hive/cluster info, harvest date, lab results, certification and movement history — while AI flags hive risk and forecasts yield, and a QR scan lets any consumer verify where their honey has been.

# Proposed Idea

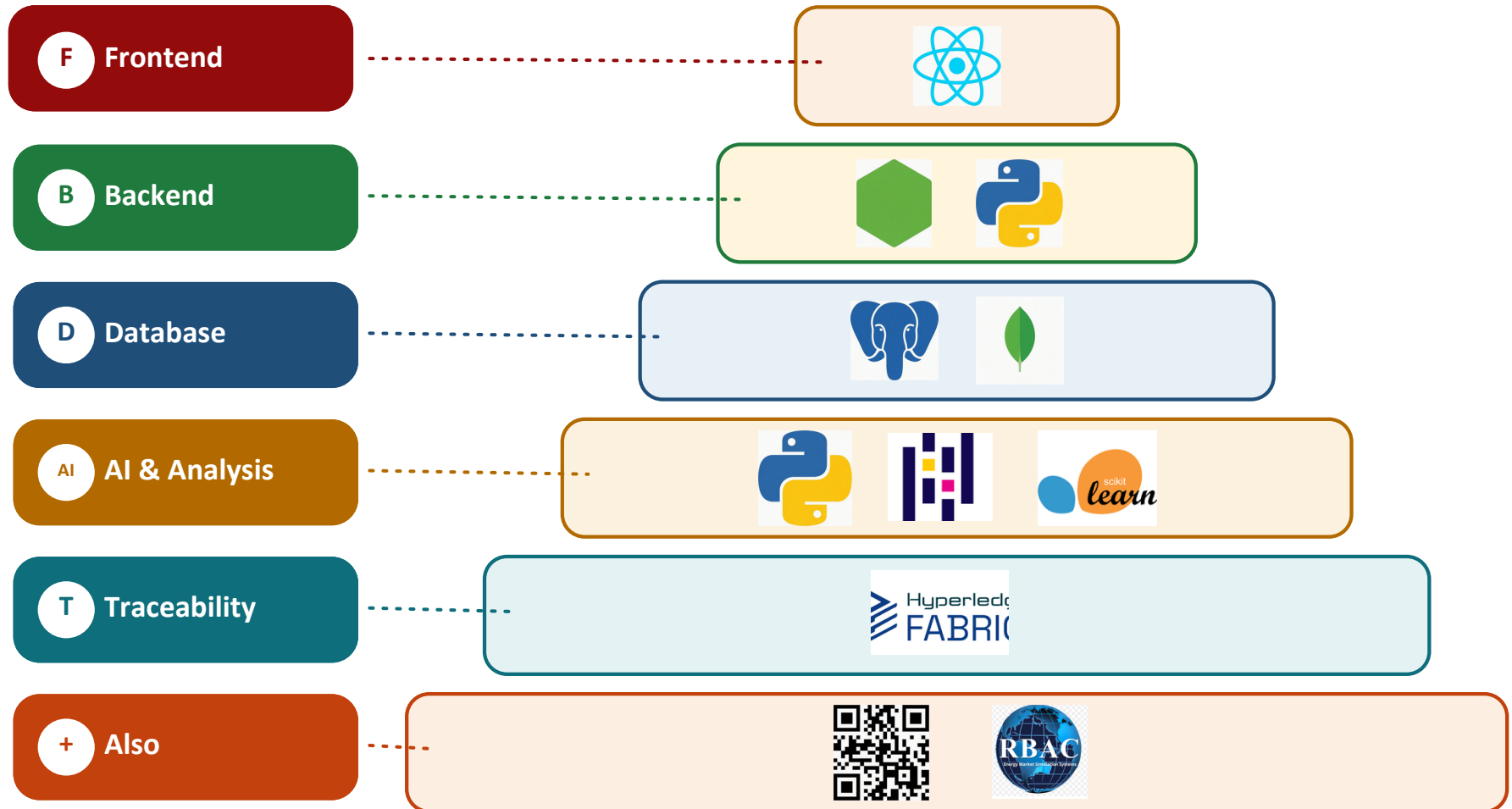
## System Architecture — BeeTrust AI



### HOW IT FLOWS



# Tech stack / Technology



# Innovation / Differentiation



**One digital record per batch**  
instead of information  
scattered across separate  
systems.



**QR-based consumer verification**  
quick access to a batch's  
verified information.



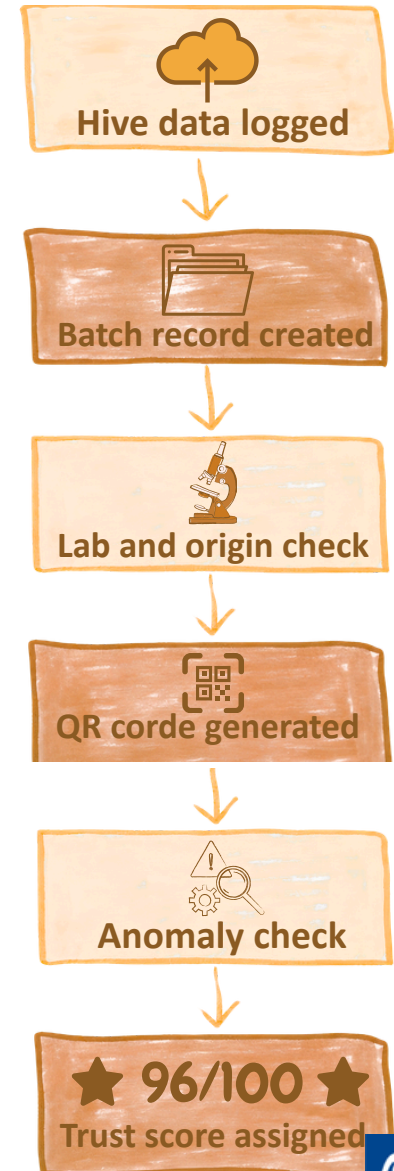
**AI-assisted hive analysis**  
historical and recorded  
data surfaces risk and  
trend.



**Supply-chain anomaly detection**  
flags gaps between  
production, batch and  
movement records.



**Production forecasting**  
past records help estimate what's coming.



# Feasibility & Expected Impact

## WHY IT IS FEASIBLE

- **Primarily software-based** solution
- **No physical sensors** required for the prototype
- Hive information entered **manually** or generated as realistic test data
- AI models trained and tested on **historical or simulated datasets**
- Real **IoT sensors** can be integrated later if required

## WHO BENEFITS

### Beekeepers

- Better **record keeping**
  - **Early risk** indicators
  - **Production estimates**
  - Better visibility of their products
- ### KVIC / Organizations
- Visibility of **honey production**
  - Easier **batch tracking**
  - Improved **procurement planning**
  - Identification of **unusual records**

## IMPACT, SCALE & TRUST

### Laboratories

- Test results **digitally linked** to specific batches
- Easier access to **batch information**

### Consumers

- Quick access to **product information**
- Clear view of **honey origin and testing**
- Increased **confidence** in verified products

### Long-Term Impact

BeeTrust AI can help create a more organized and transparent honey ecosystem while supporting genuine beekeepers and improving consumer confidence.

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# Team composition

| Member no | Skills                                       |
|-----------|----------------------------------------------|
| Member 1  | Team lead, backend (Python)                  |
| Member 2  | Frontend development (React.js)              |
| Member 3  | AI/ML development (Python, Scikit-learn)     |
| Member 4  | Traceability & security (Hyperledger Fabric) |
| Member 5  | Database & APIs (PostgreSQL / MongoDB)       |
| Member 6  | Testing & documentation                      |